Lecture 08

Multiple Market Failures and Second-Best Policies

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ECON 467

Roadmap

1. What happens when we have another market failure like market power?
2. How do second-best policies like output taxes or intensity standards work?

Market power and pollution

Market power

Lets consider two extreme cases to understand whether and how market power matters

1. Perfect competition
2. Monopoly

Market power

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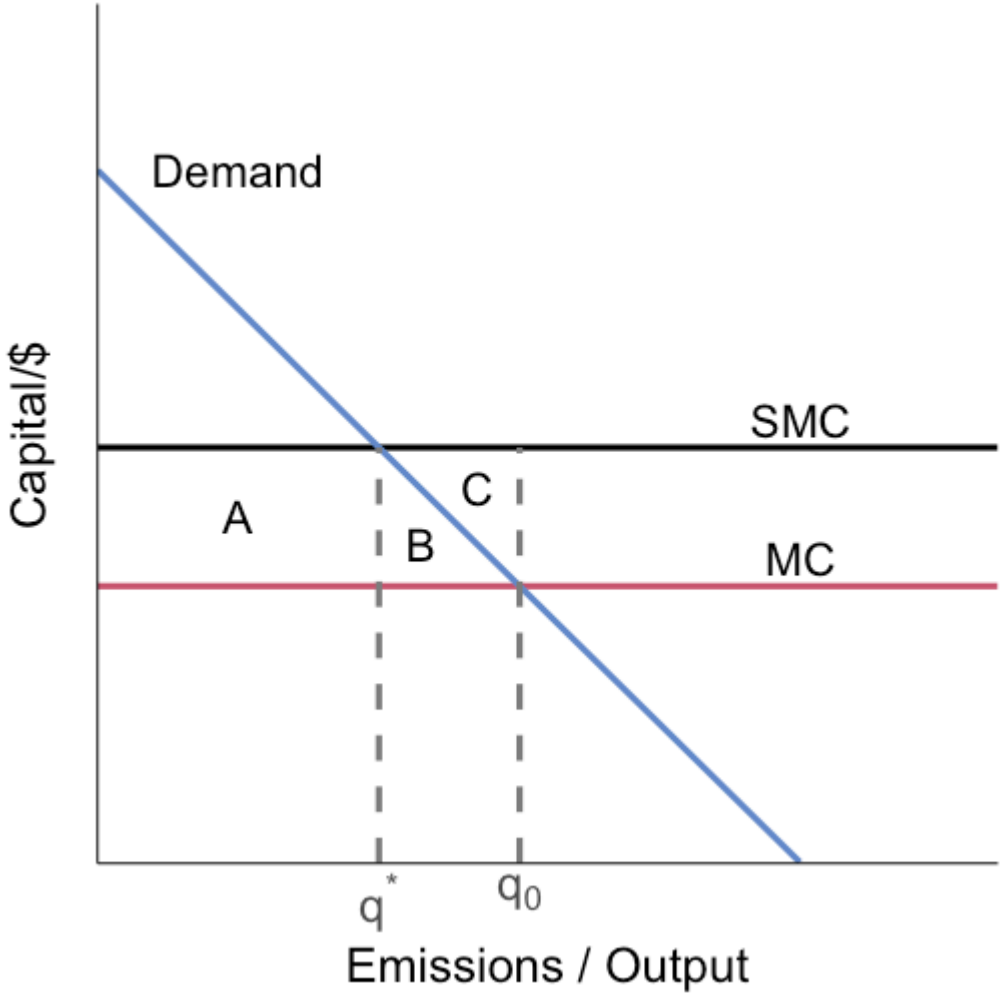
1. Perfect competition
2. Monopoly

In both cases we will assume that:

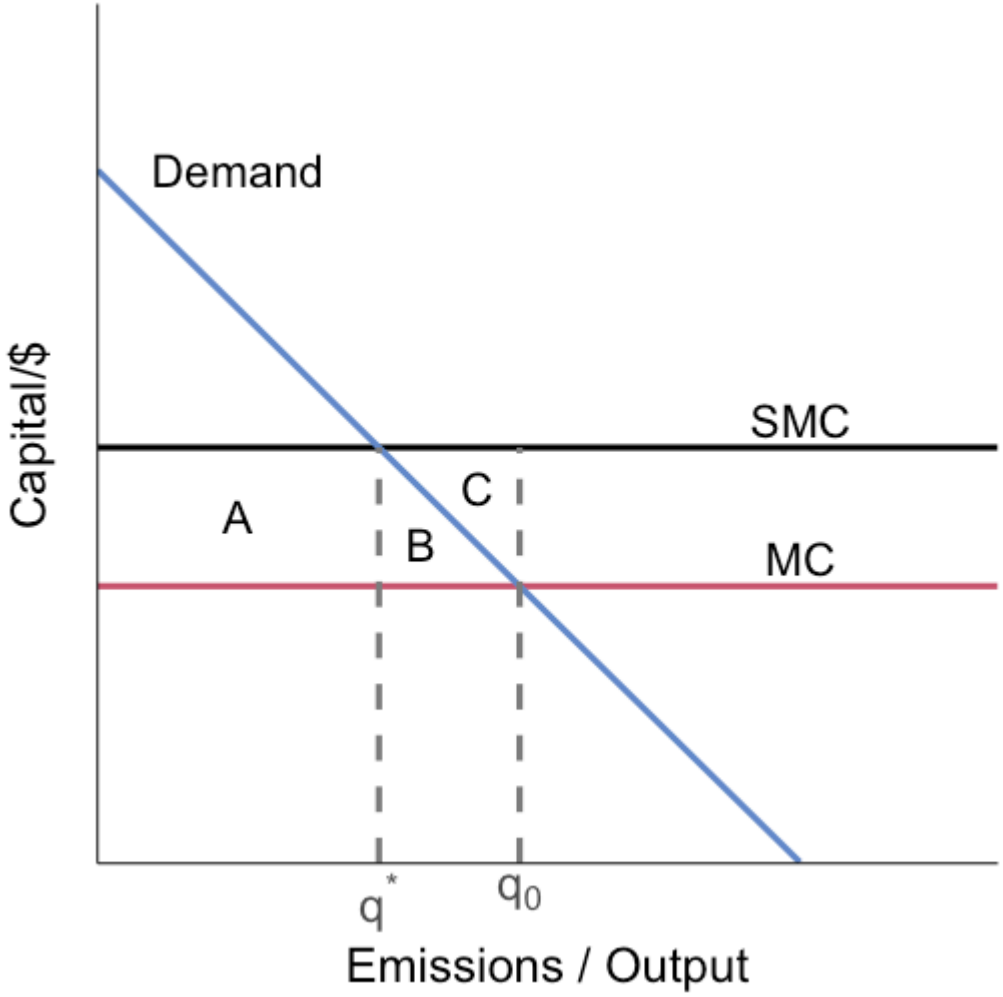
1. Marginal costs of production are constant MC
2. Emissions are perfectly proportional to output, so each additional unit of output creates the same marginal damage and gives us constant social marginal cost $SMC = MC + MD$

Perfect competition



The effect of moving from $q_0 \rightarrow q^*$ using a tax equal to marginal damage (SMC - MC):

Perfect competition



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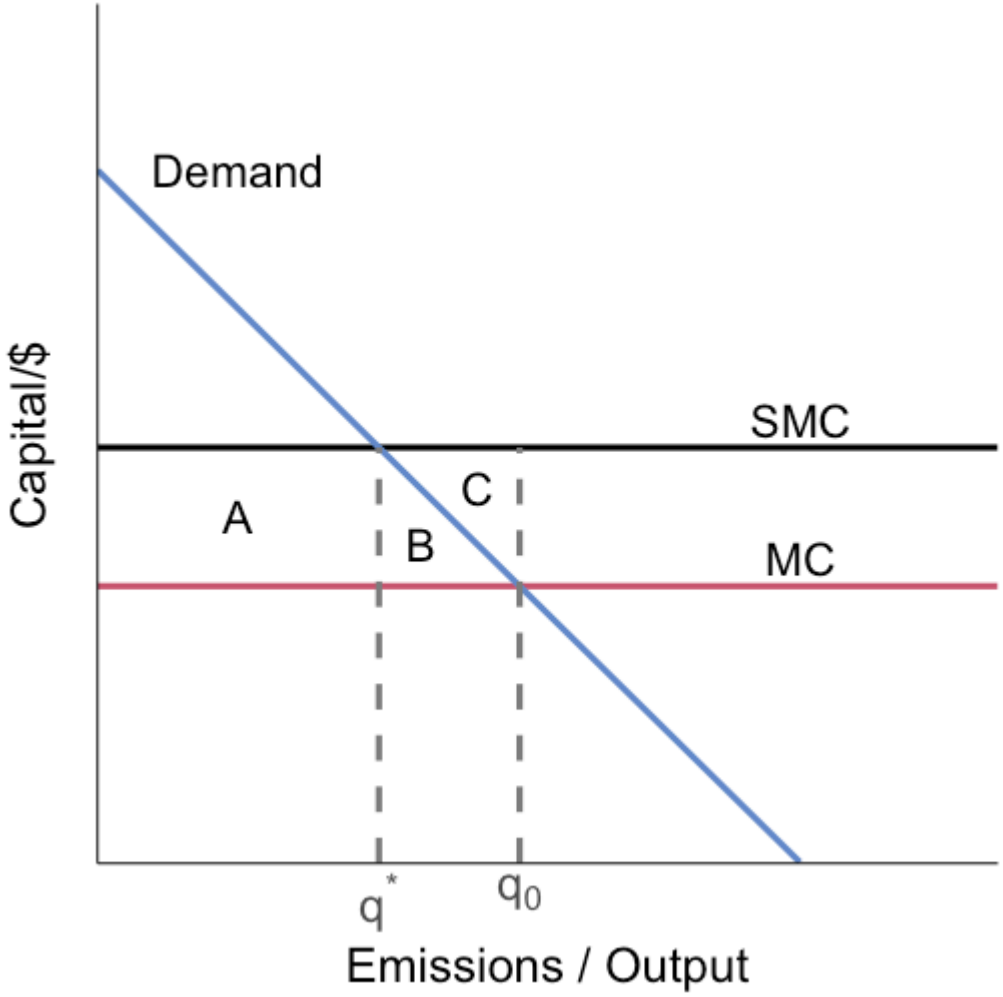

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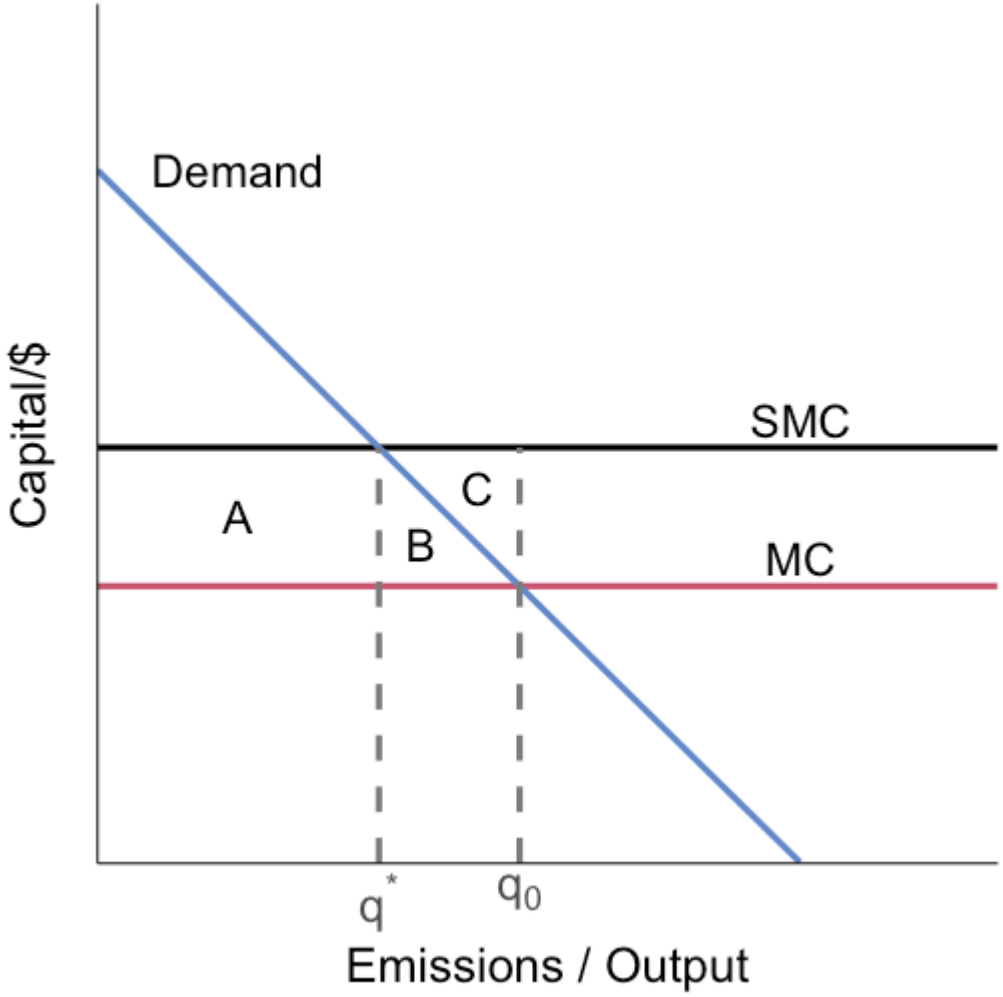

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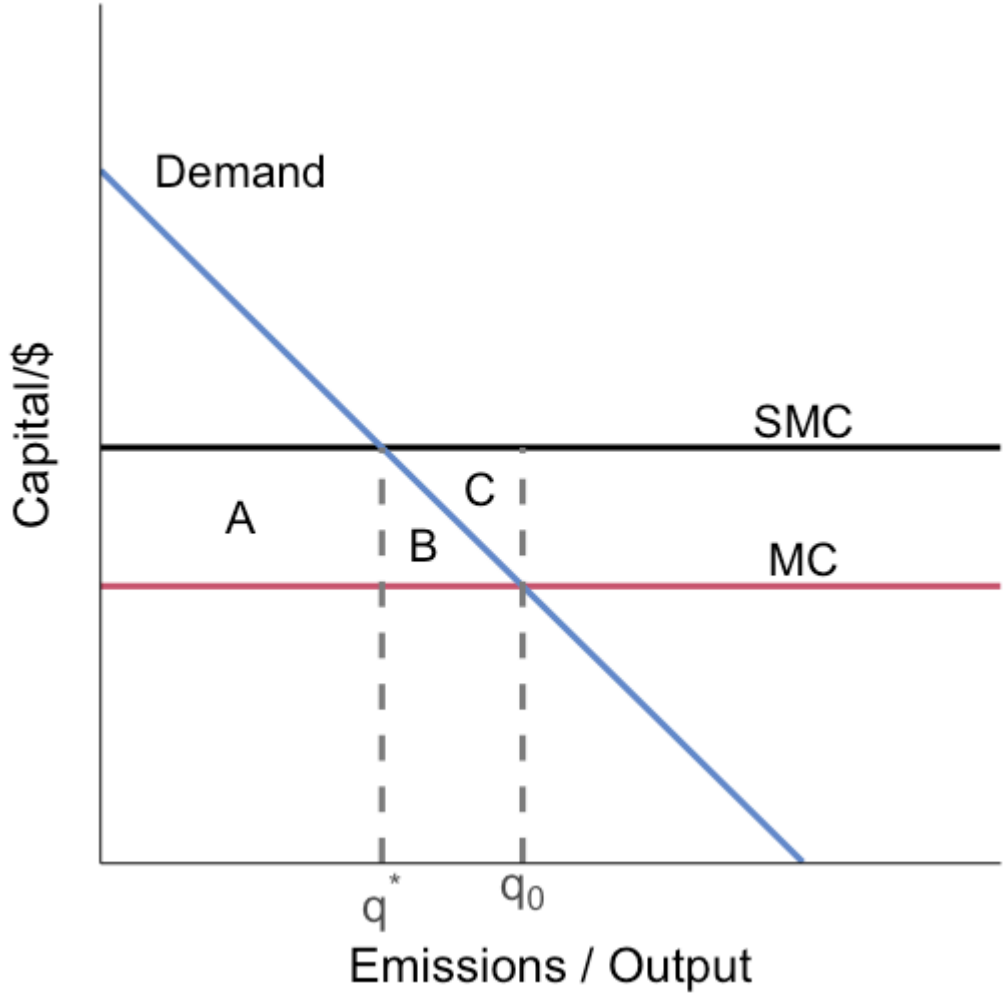

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Loss in CS: -(A+B)

Avoided damages: B+C

Tax revenue: A

Net gain: -(A+B) + (B+C) + A = C

Monopoly

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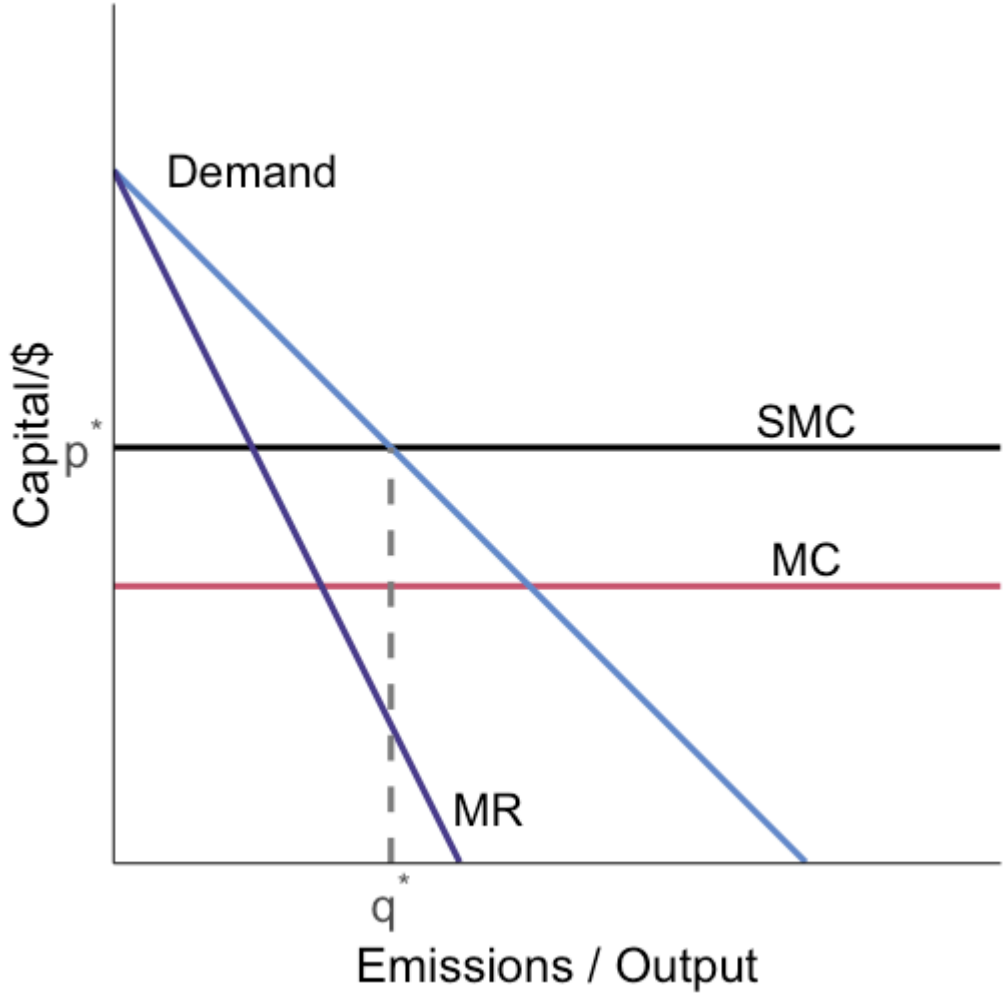
Why?

The monopolist accounts for how additional output lowers the market price on inframarginal units

Monopoly

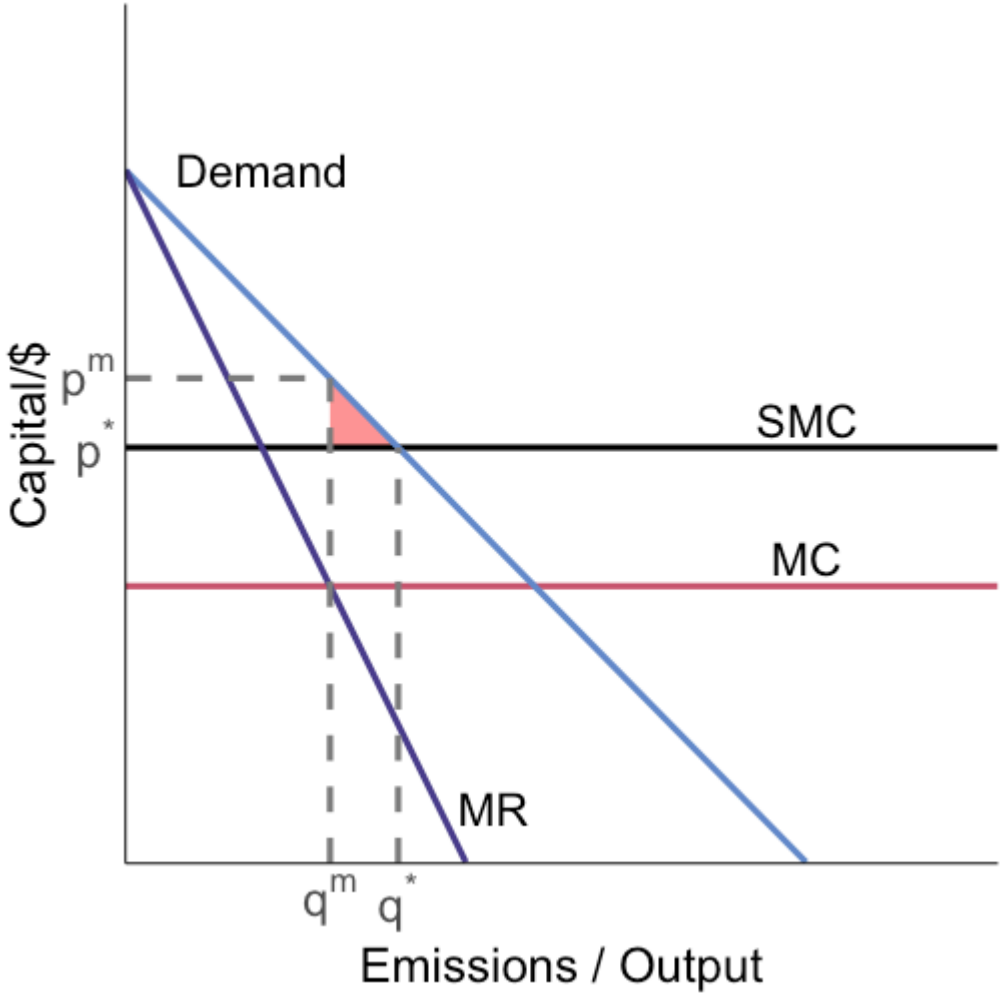


The socially efficient allocation is where social marginal cost is equal to the social marginal benefit

This is where SMC crosses the demand curve: (q^*, p^*)

What is the welfare outcome under the unregulated monopolist outcome?

Monopoly

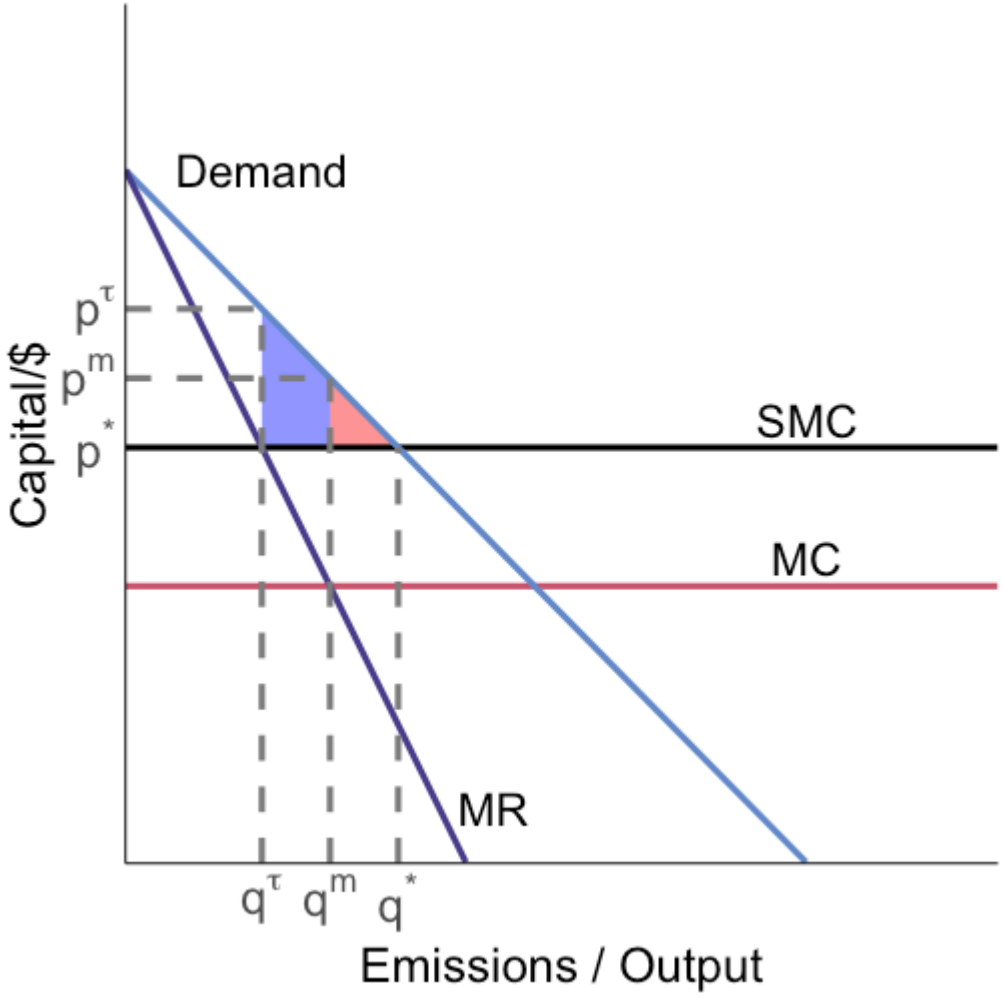


In the absence of regulation, the monopolist maximizes profit where MR = MC: (q^m, p^m)

This results in deadweight loss equal to the **red** area

Now what happens if we set a Pigouvian tax equal to marginal damage?

Monopoly

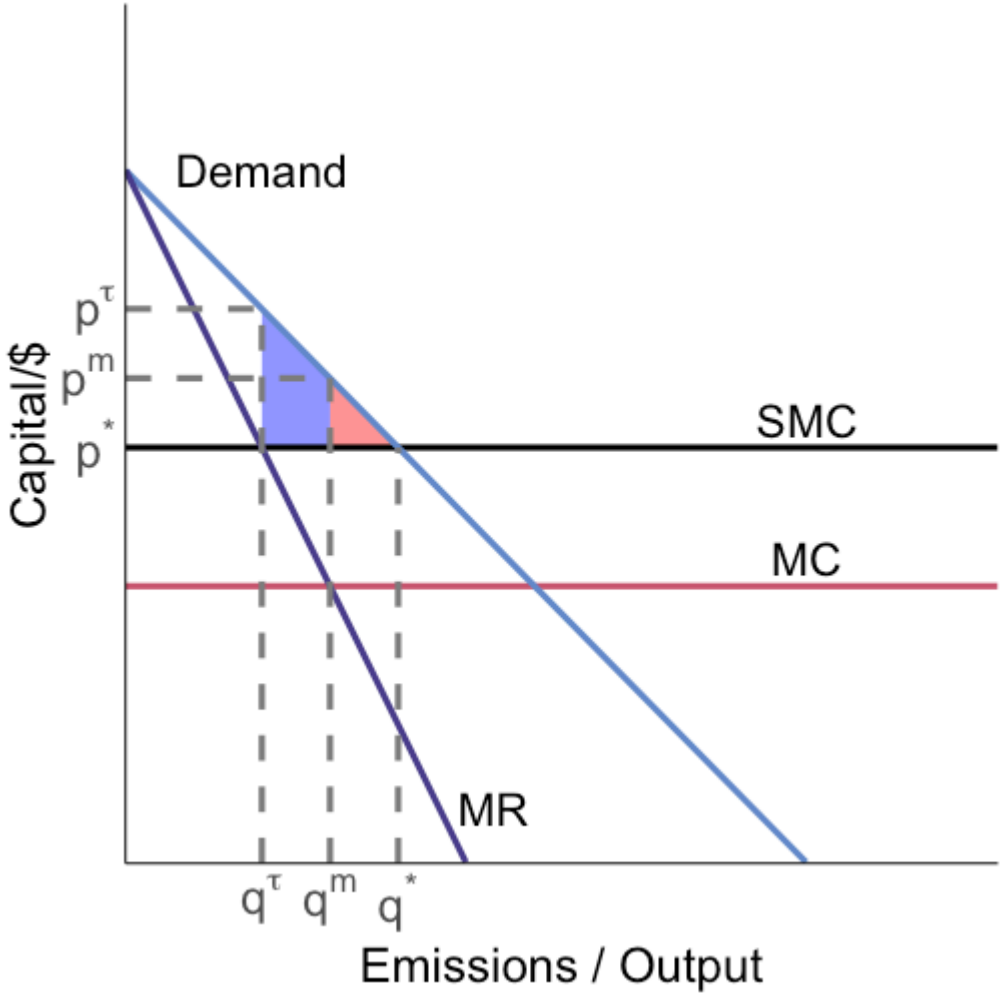


The Pigouvian tax restricts output even more, adding deadweight loss equal to the **blue** area on top of the deadweight loss in the **red** area

The tax actually made us worse off by the blue area!

Why?

Monopoly



We now have two market failures:

1. Market power
2. Pollution externality

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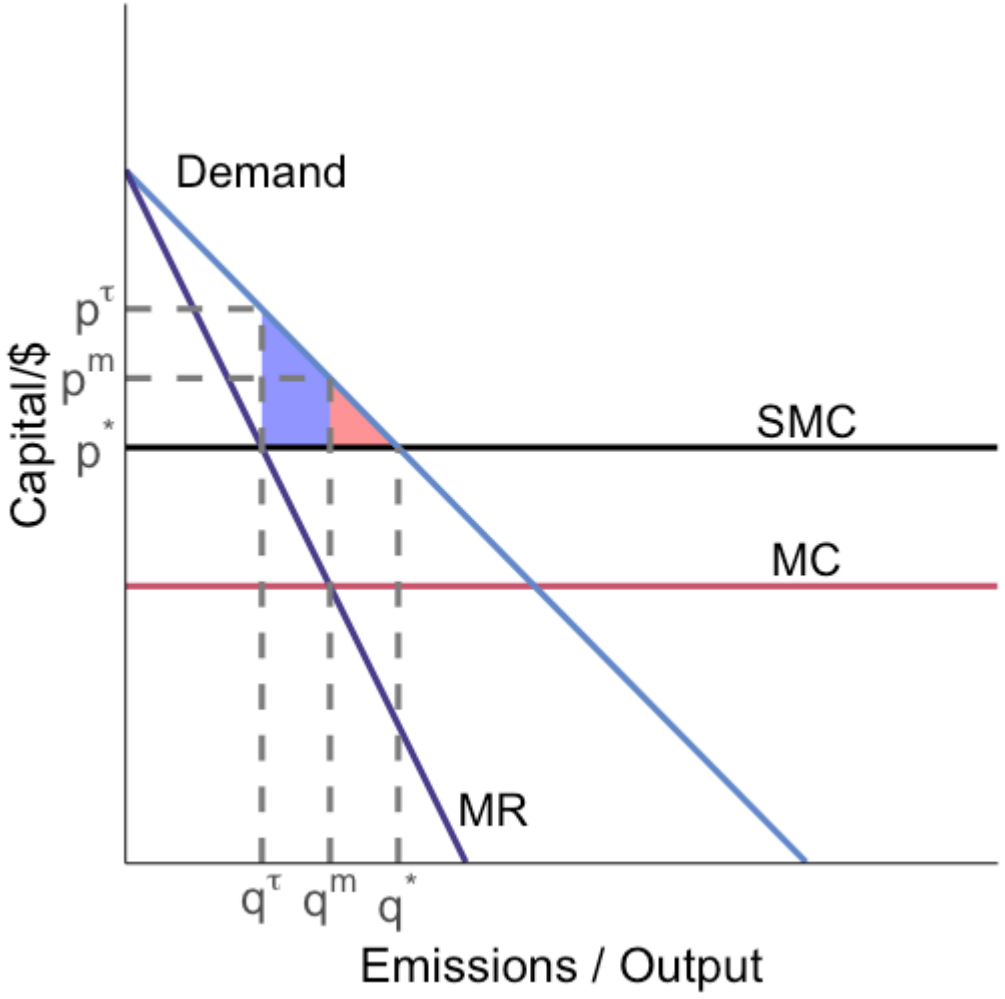

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With market power, the unregulated equilibrium quantity is **too low**

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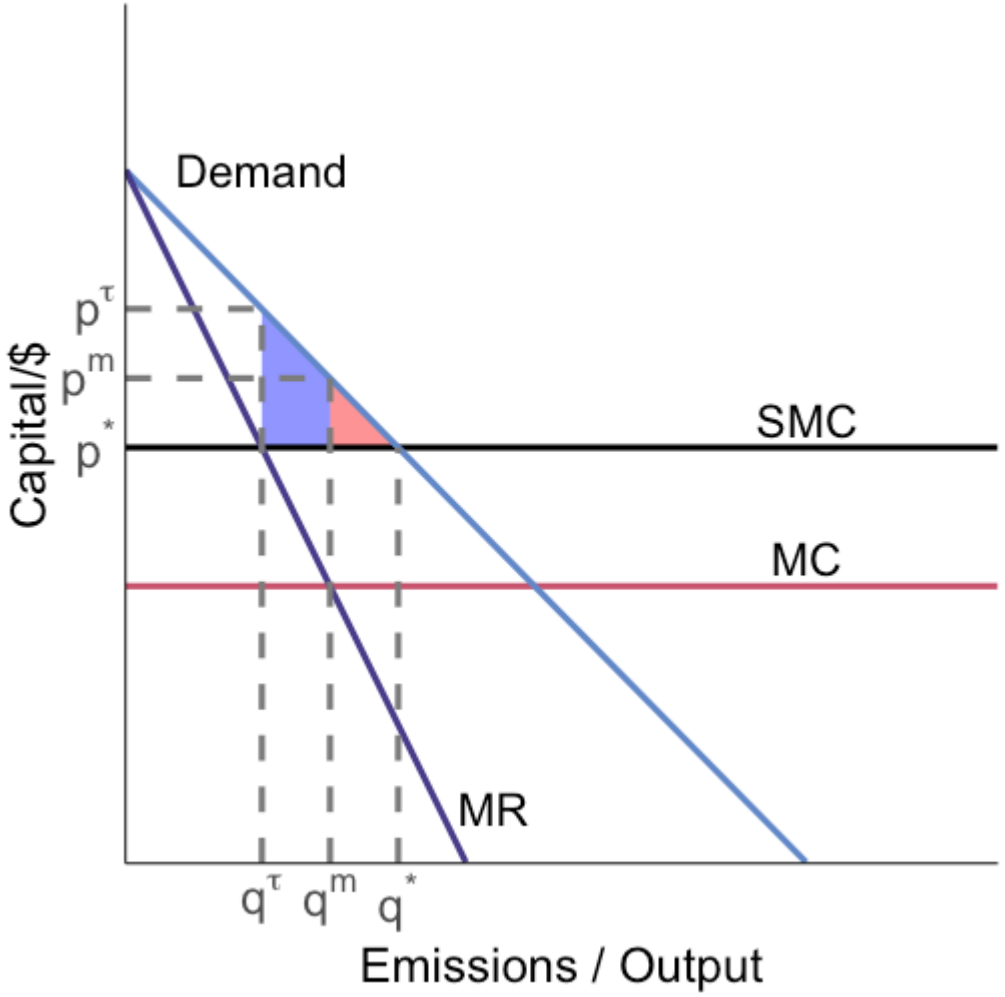

We now have two market failures:

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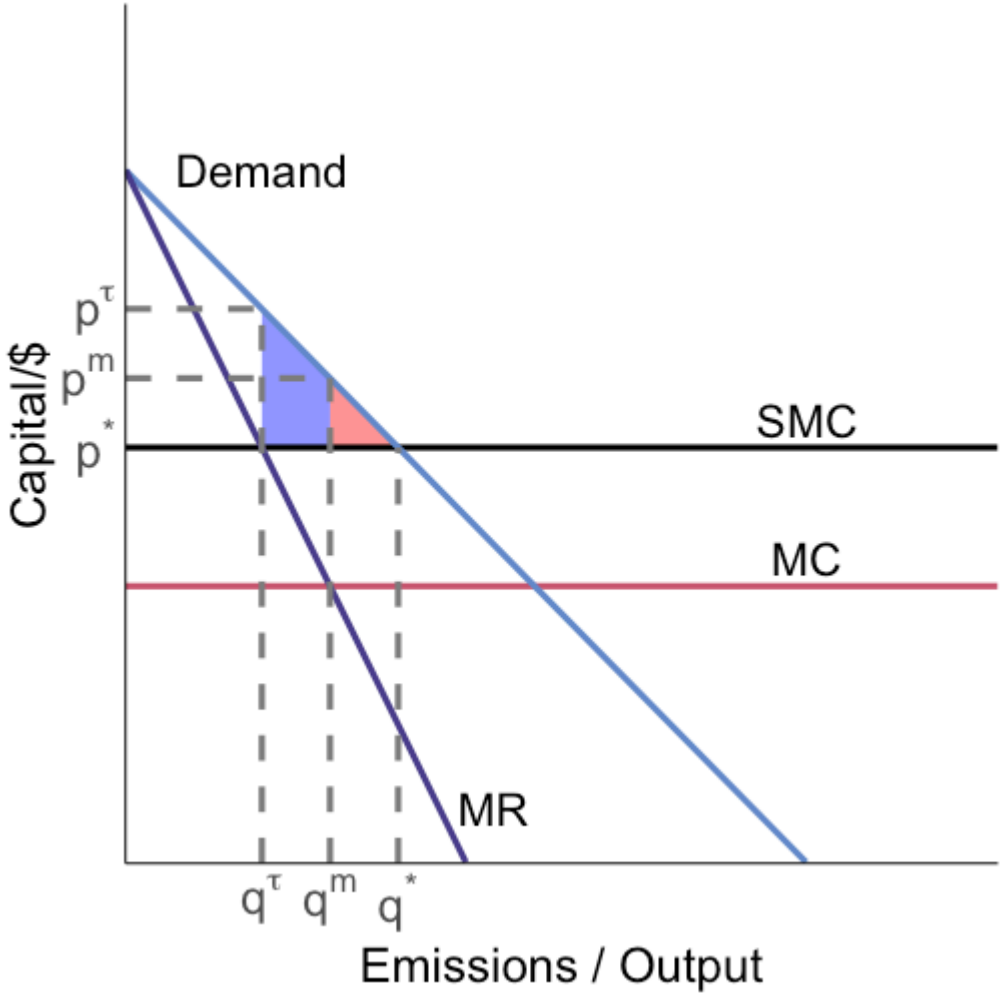
With market power, the unregulated equilibrium quantity is **too low**

With a pollution externality, the unregulated equilibrium quantity is **too high**

Monopoly



They have opposing forces on quantities, so the market failures offset each other (partially)

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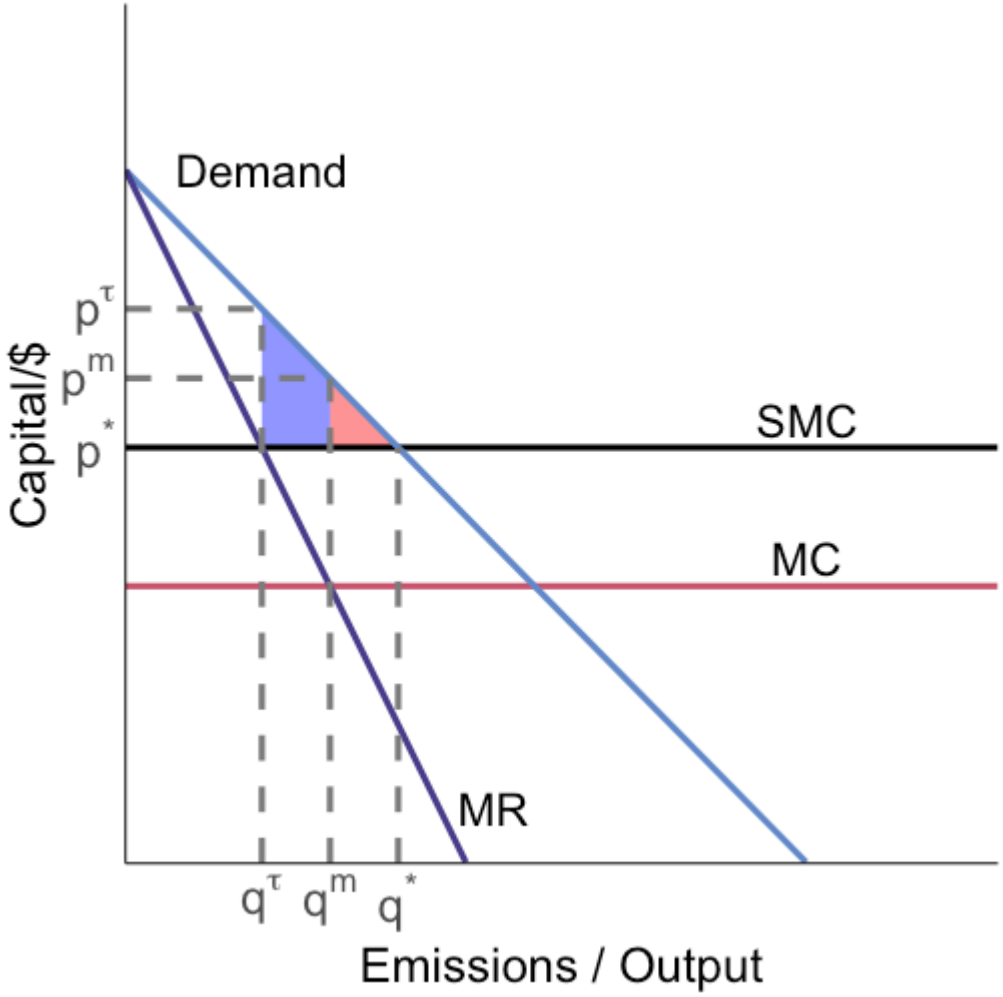

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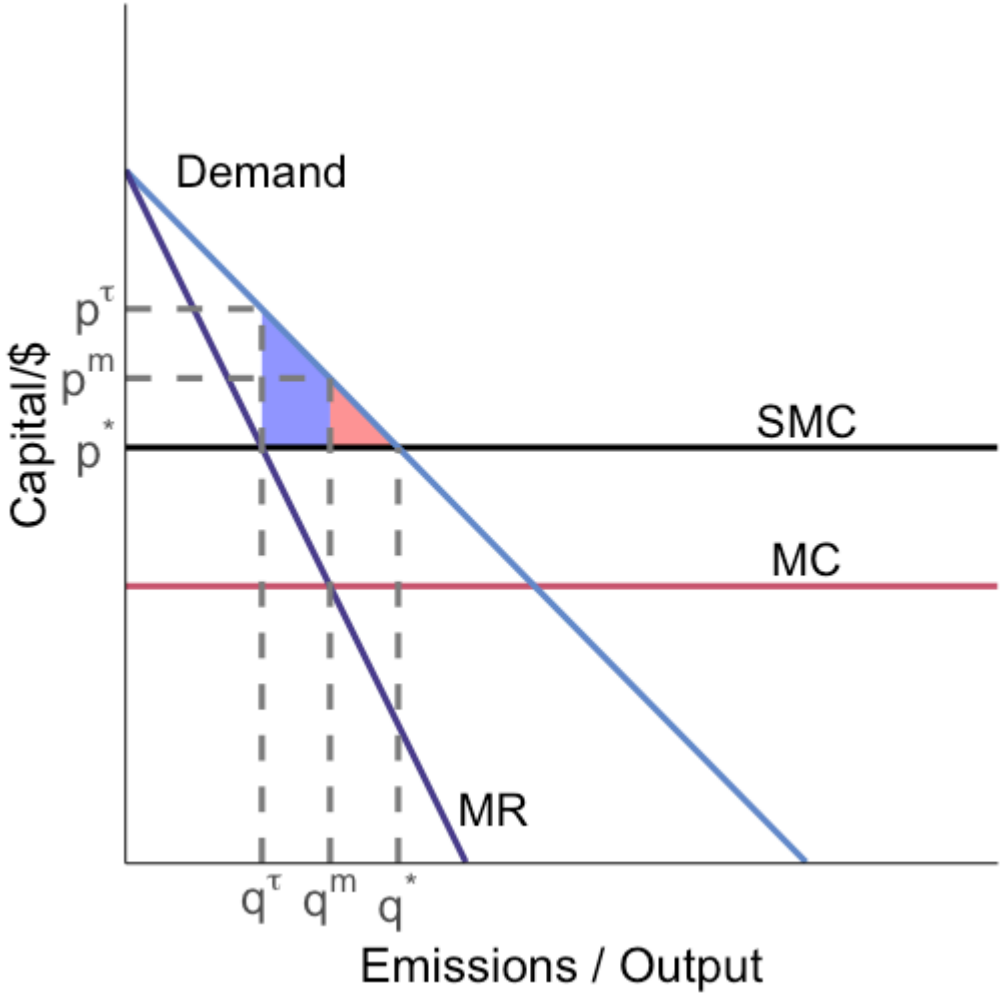
This means that if we fully correct the pollution externality, we no longer get the off-setting benefit and have the full welfare cost of market power

Monopoly



What is the actual optimal thing to do here?

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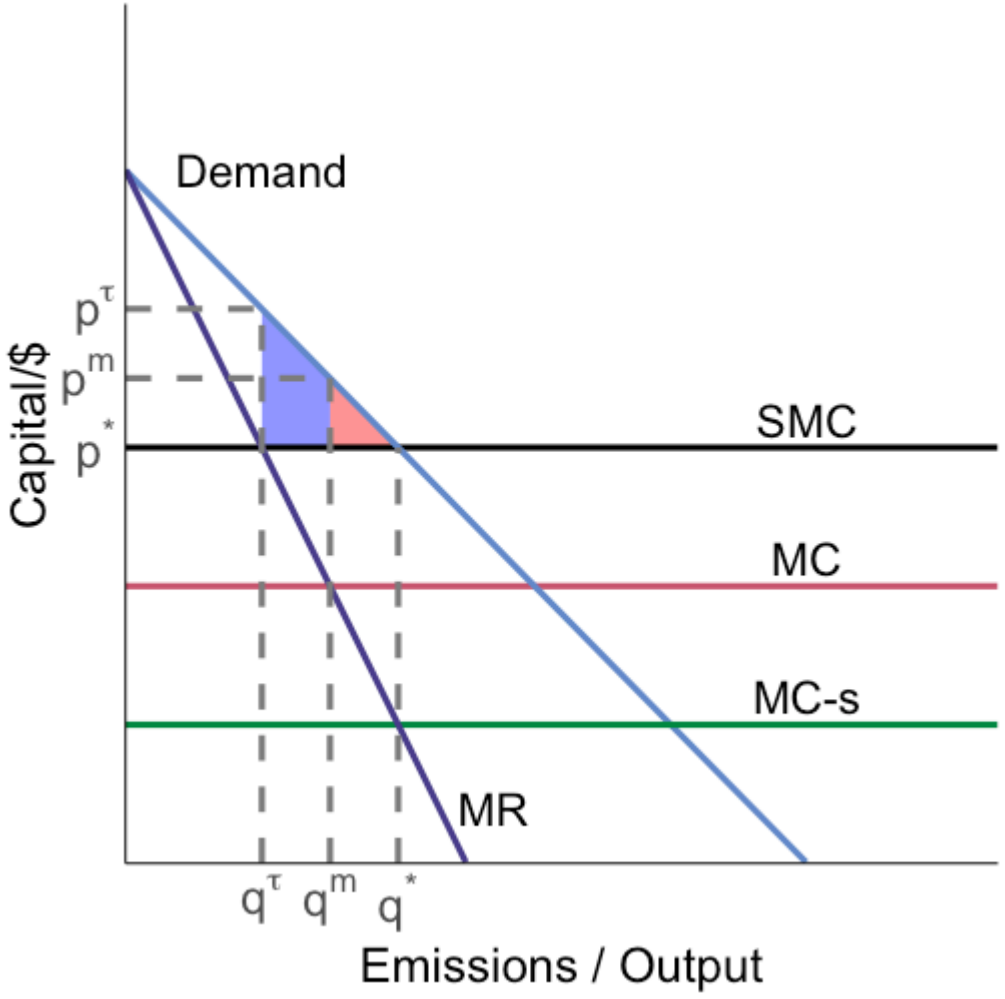

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subsidize output at rate s so $MC - s$ crosses MR at q^*

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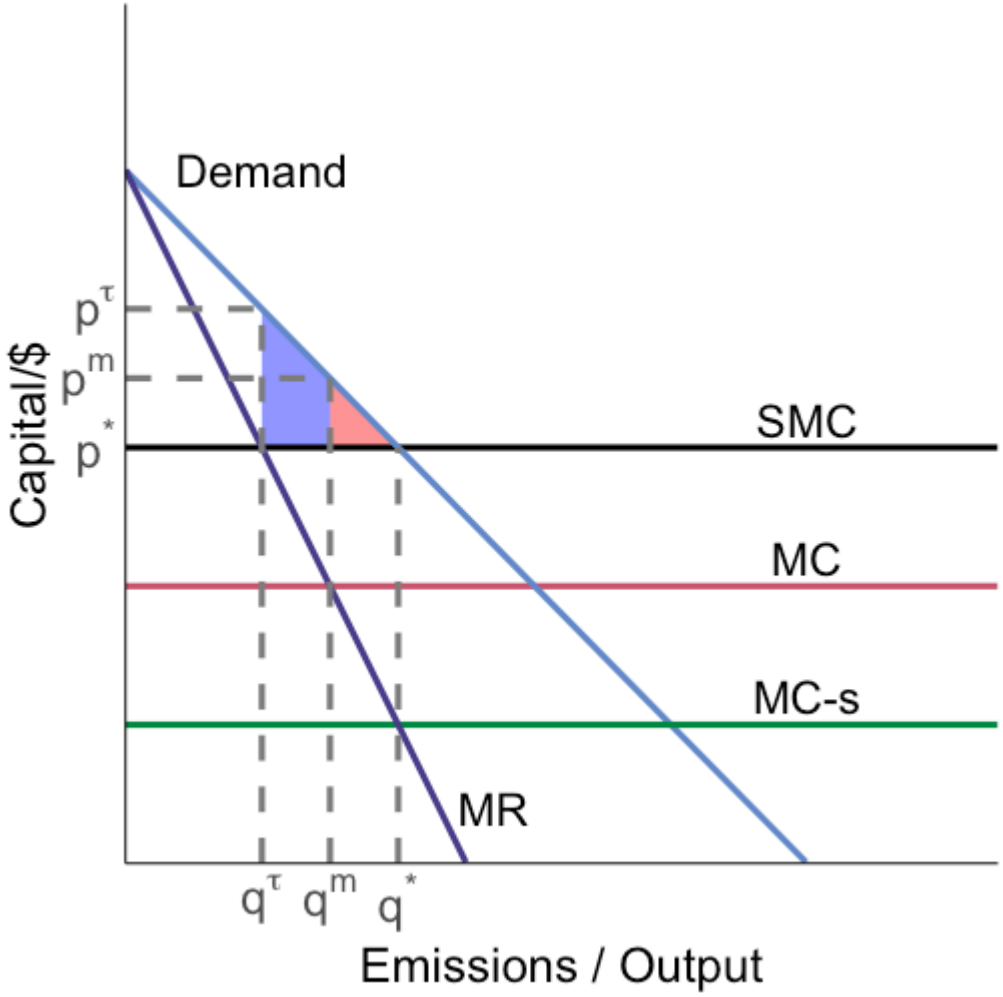

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In this example, the market power externality dominates the pollution externality: we need to increase output

Monopoly

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You just need marginal damages to be sufficiently large relative to the market power effect on quantity

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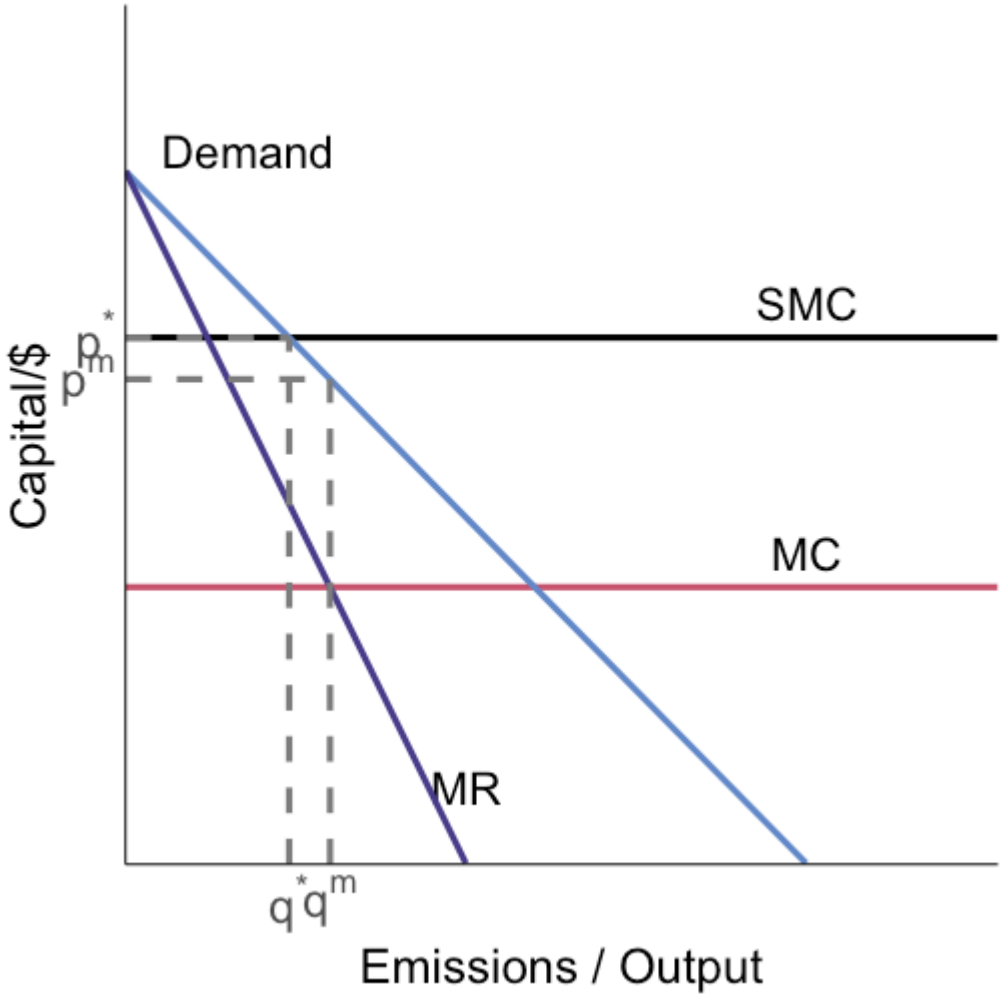

What if marginal damages are larger?

Suppose SMC is high enough that demand and SMC intersect to the **left** of q^m

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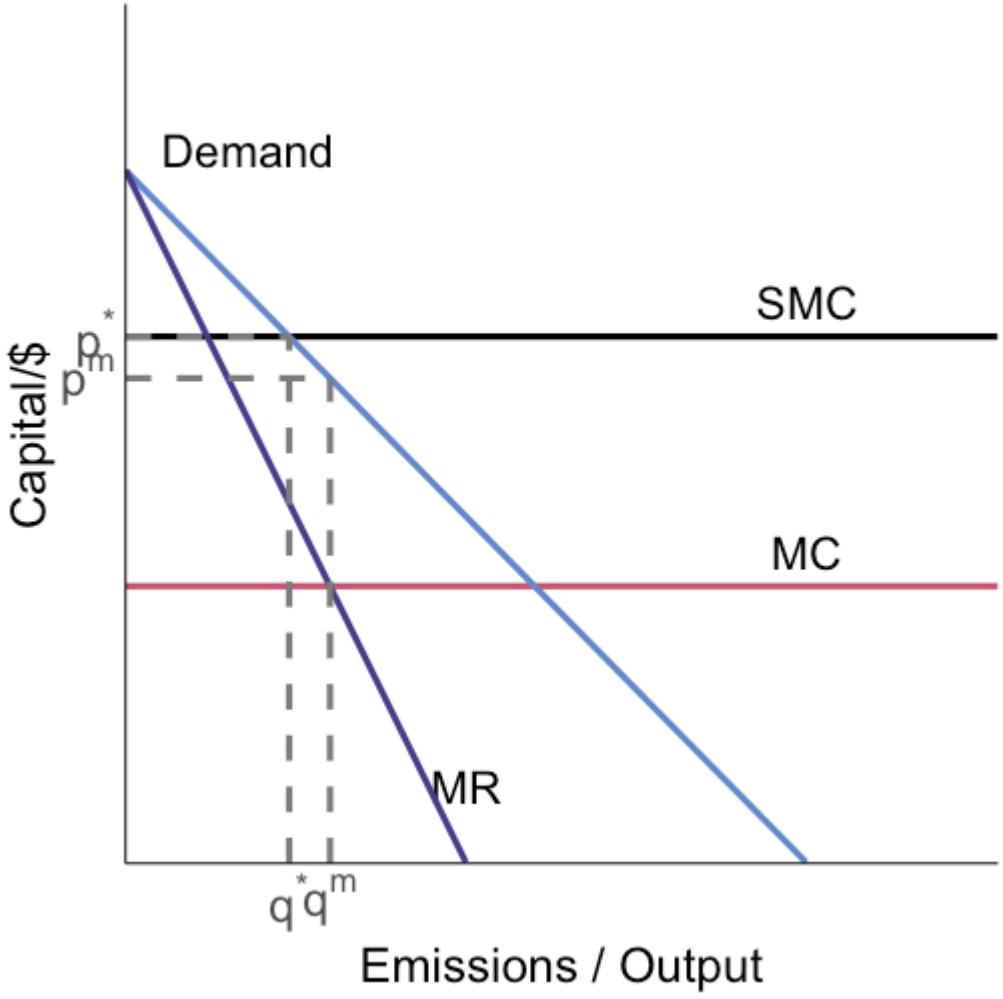

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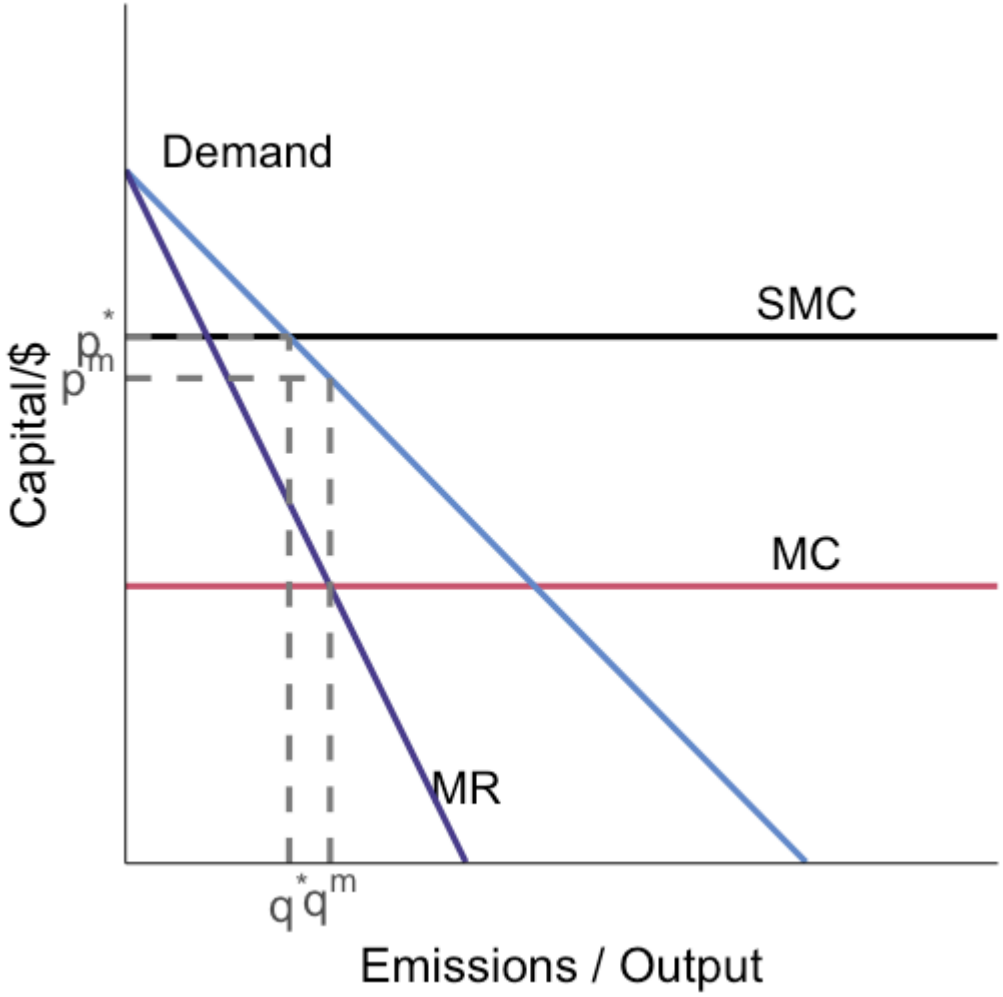

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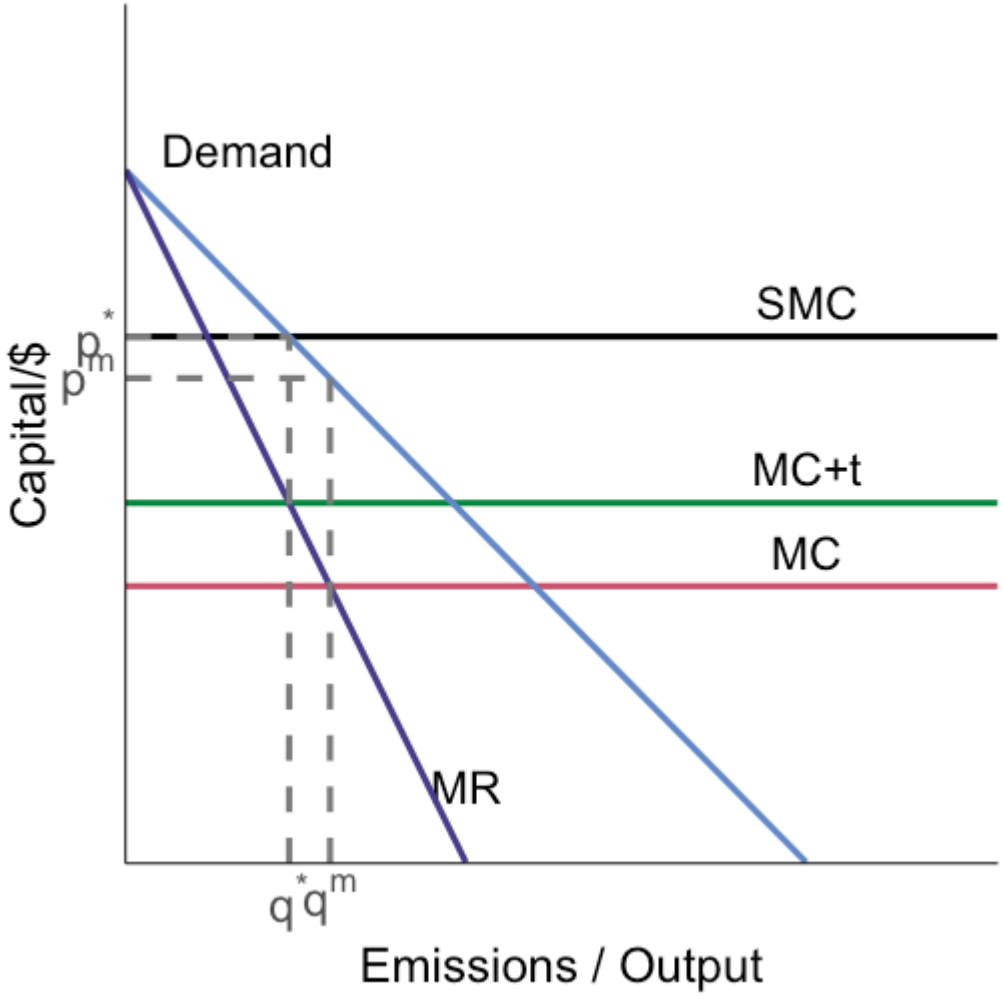

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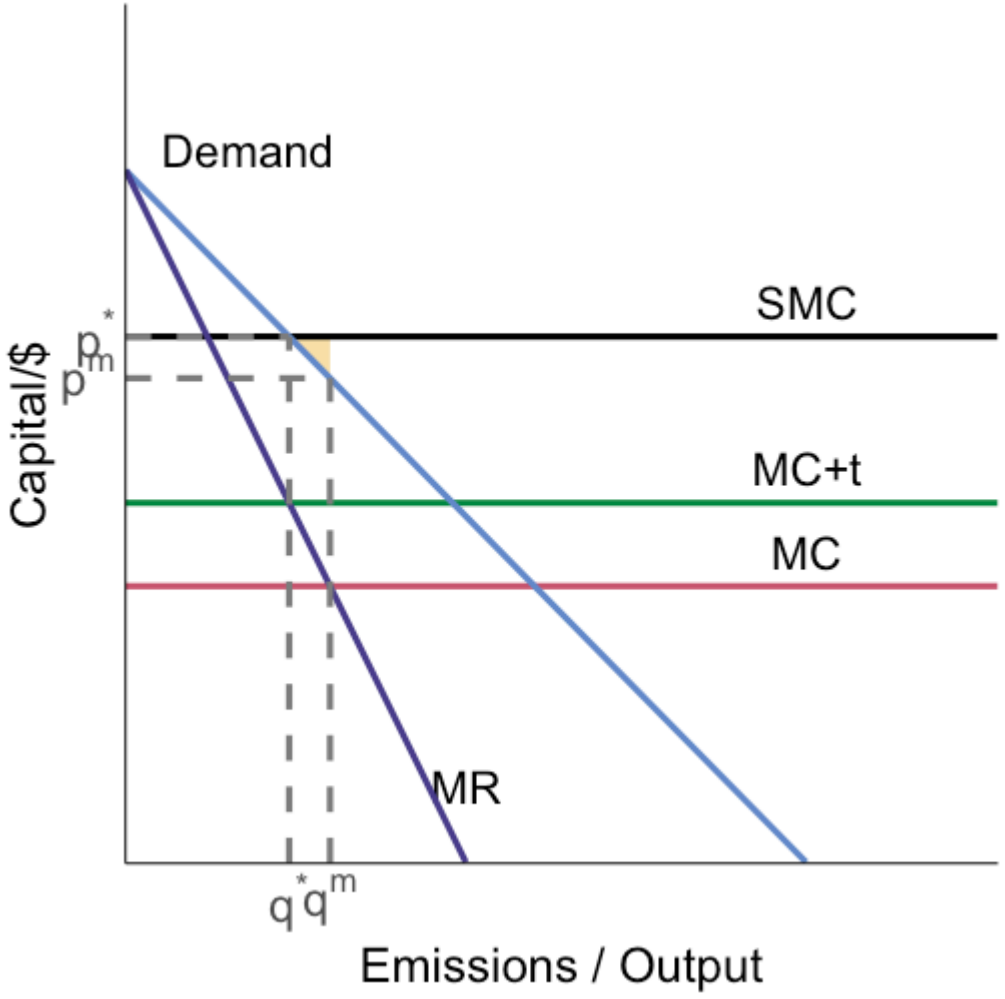

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Welfare gain from this correction is the reduction in overproduction between q^m and q^*

Monopoly: more generally

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What changes is we can no longer fix them both with just a pollution tax/subsidy

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What does that mean here?

We need:

1. Pollution tax
2. Output subsidy

The tax incentivizes abatement, the subsidy incentivizes production

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

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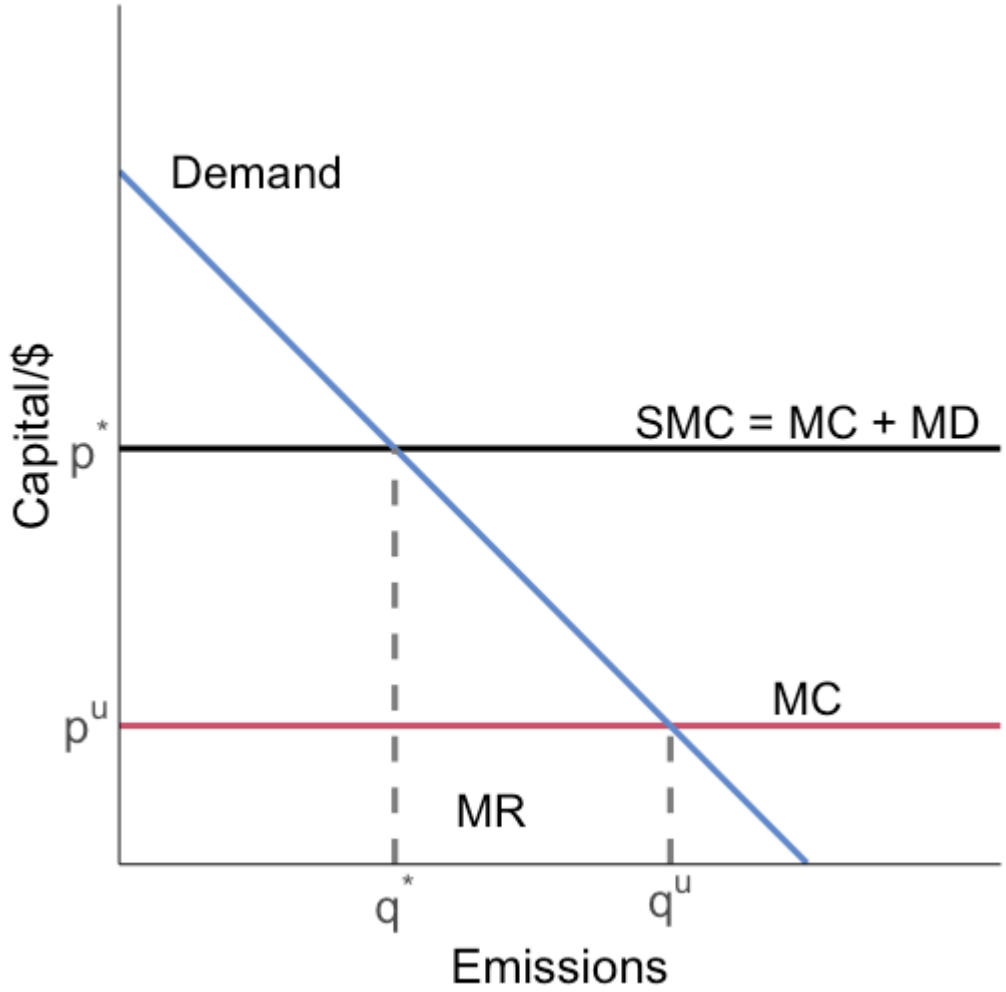
Will this be efficient?

If so, what assumptions do we need about the production process?

Output taxes



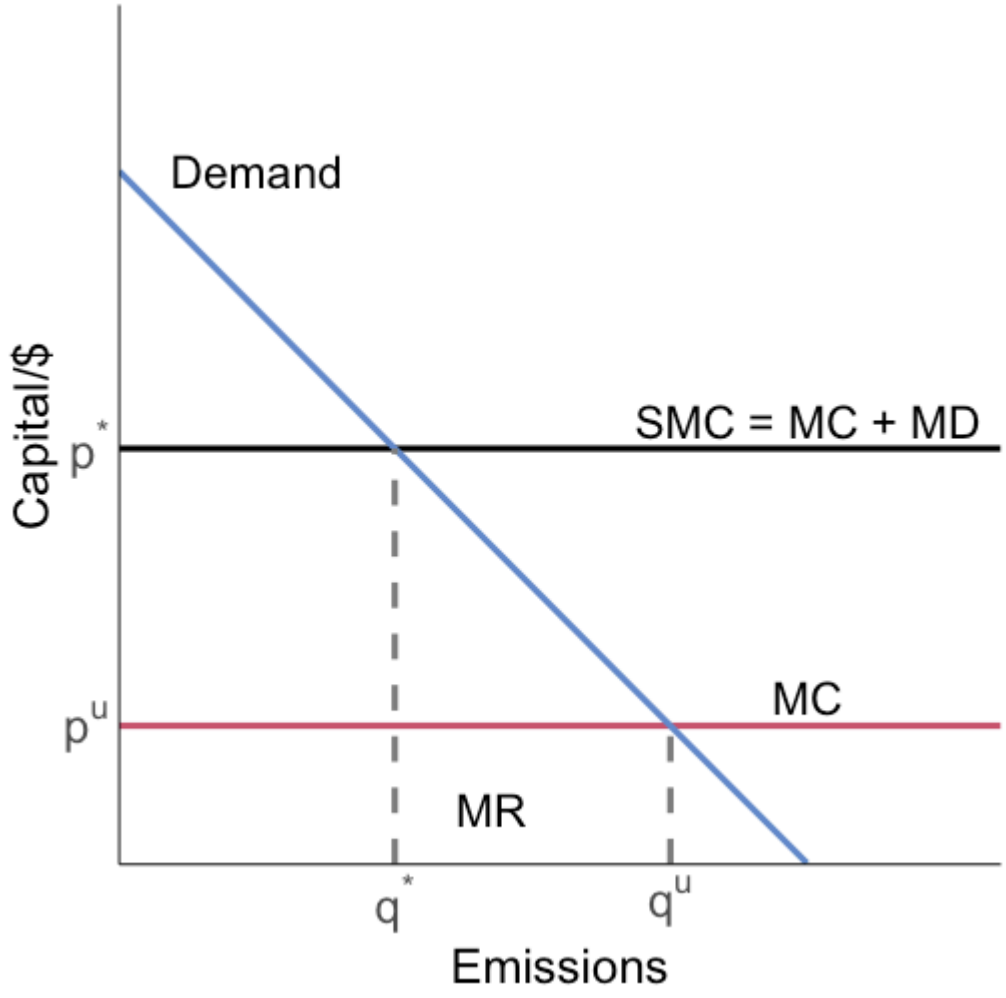
Assume emissions are proportional to output

And MD is constant

The firm chooses to produce/emit at q^u in the unregulated equilibrium

If we tax output equal to MD we can achieve the socially optimal allocation q^*

Output taxes



An output tax can be efficient, **if we assume that emissions are proportional to output**

Now let's break the link between output and emissions by writing down a slightly more complicated model where the firm chooses emissions and output separately

Output taxes, part two

Here's our model:

- General production function using labor and emissions as inputs: $Q = f(L, E)$
- The firm pays wages w to labor and rental rate r for the emissions-generating input E
- The firm receives a price p per unit of output
- Emissions cause marginal damage d

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What does an output tax τ_o do versus a regular emission tax τ_e?

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The regulator wants the firm to internalize its social costs:

$$\max_{L,E} pf(L,E) - wL - rE - dE$$

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For a social optimum, labor is priced at its private cost, but emissions must face the extra damage term d

Output taxes, part two

The firm's problem for the output tax is:

$$\max_{L,E}(p - \tau_o)f(L, E) - wL - rE$$

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What do these conditions mean?

Output taxes, part two

The output tax reduces the value of the marginal product of **both** inputs

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$$p\frac{\partial f(L,E)}{\partial L} = w + \tau_o \frac{\partial f(L,E)}{\partial L}$$

Output taxes, part two

The output tax reduces the value of the marginal product of **both** inputs

Rearranging the labor condition gives:

$$p\frac{\partial f(L,E)}{\partial L} = w + \tau_o \frac{\partial f(L,E)}{\partial L}$$

So the output tax implicitly taxes clean labor too, which is **not efficient**

Output taxes, part two

How does this compare to a pure emission tax?

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A tax of $\tau_e = d$ can achieve the efficient allocation!

Output taxes takeaways

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If emissions can be chosen separately from outcome by the firm, this is no longer true

In this case an output tax incorrectly taxes our clean inputs

Intensity standards

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Instead of pricing each unit of emissions directly, the regulator sets an emissions rate per unit of output

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These policies are common when direct emissions monitoring is difficult but output composition is observable

Intensity standards: RPS example

What is a Renewables Portfolio Standard (RPS)?

aka Renewable Energy/Electricity Standard (RES)

Renewables Portfolio Standard

A requirement on retail electric suppliers…
To supply a minimum percentage or amount of their retail load…
With eligible sources of renewable energy

Typically	Backed with penalties of some form
Often	Accompanied by a tradable renewable energy certificate (REC) program to facilitate compliance
Never	Designed the same in any two states

This report covers U.S. state RPS policies. It does <u>not</u> cover:

- Voluntary renewable electricity goals
- Broader clean energy requirements without a renewables-specific component (briefly discussed in a side-bar)
- RPS policies outside of the United States or in U.S. territories

Intensity standards: the constraint

Suppose we have a high-emissions product H and a lower-emissions product L

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An intensity standard can be written as:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

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The policy limits average emissions per unit of output, not total emissions directly

Intensity standards: firm incentives

The firm's choices satisfy:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

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If $\beta_i > \sigma$, the policy creates an implicit **tax** on that output

Intensity standards: firm incentives

The firm's choices satisfy:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

If $\beta_i > \sigma$, the policy creates an implicit **tax** on that output

If $\beta_i < \sigma$, the policy creates an implicit **subsidy** on that output

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

So an intensity standard is equivalent to:

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This is exactly why it can perform like a second-best policy

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That is the core problem: we generally do **not** want to subsidize positive emissions

Why this is second-best

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So it can improve on no policy, but usually cannot hit the first-best allocation

Intensity standards: scale vs composition

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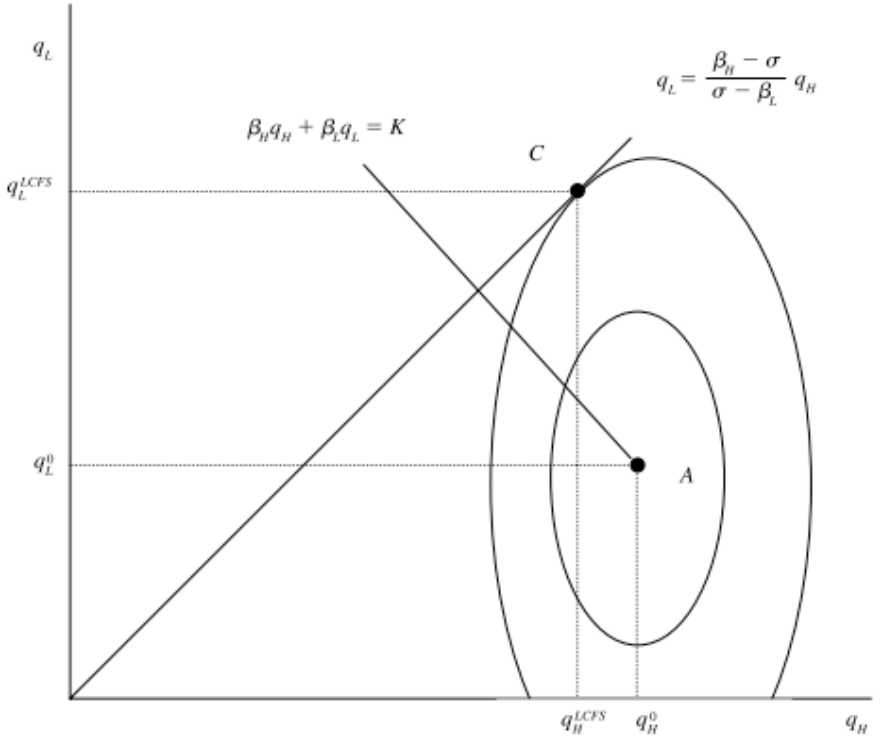
So total emissions can rise if the scale effect dominates the composition effect

Intensity standards: perverse incentives

How to read this figure:

1. The dashed ovals are iso-profit curves: each oval is a set of (Q_H, Q_L) choices with the same private profit
2. The intensity-standard line is the policy constraint:

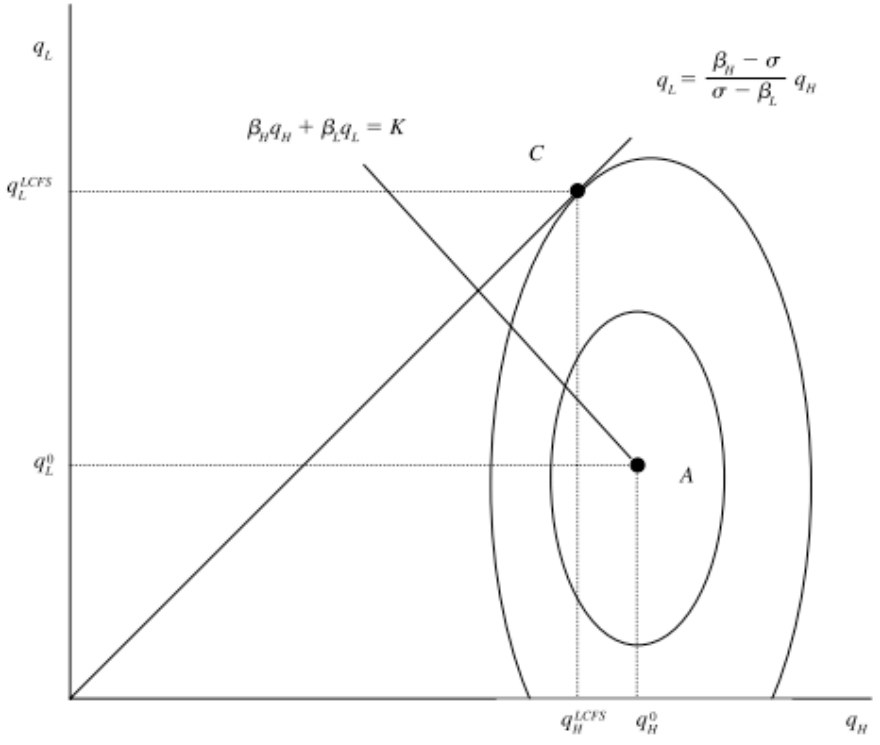
$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$



Intensity standards: perverse incentives

How to read this figure:

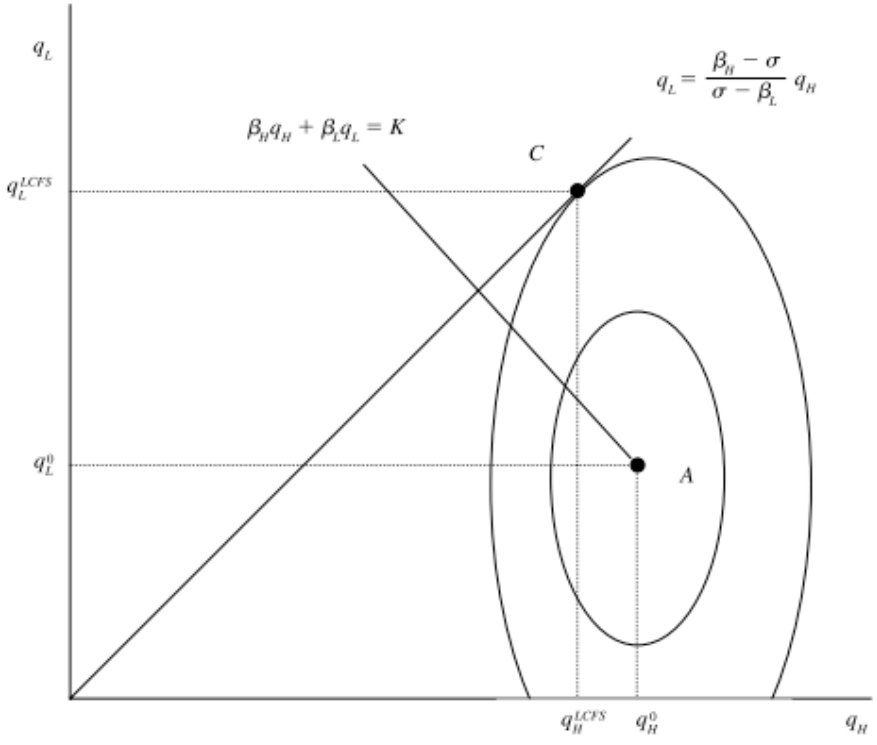
1. Point A is the unregulated profit-maximizing choice
2. The line through A labeled K shows choices with the same total emissions as A
3. Points up and to the right of K imply higher total emissions



Intensity standards: perverse incentives

What the standard does:

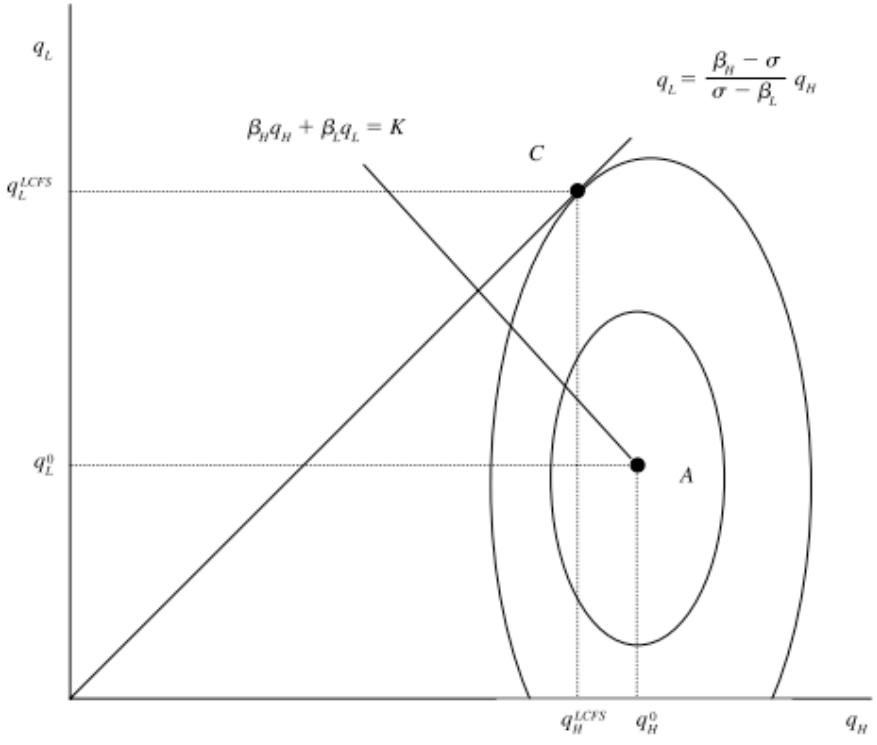
1. The intensity standard forces choices onto the intensity line
2. The firm moves from A to C, where the highest attainable iso-profit oval touches that line



Intensity standards: intuition

What the standard does:

1. Relative to A, the firm cuts dirty output H, but expands cleaner output L
2. Emissions intensity falls by construction, but total output can still rise

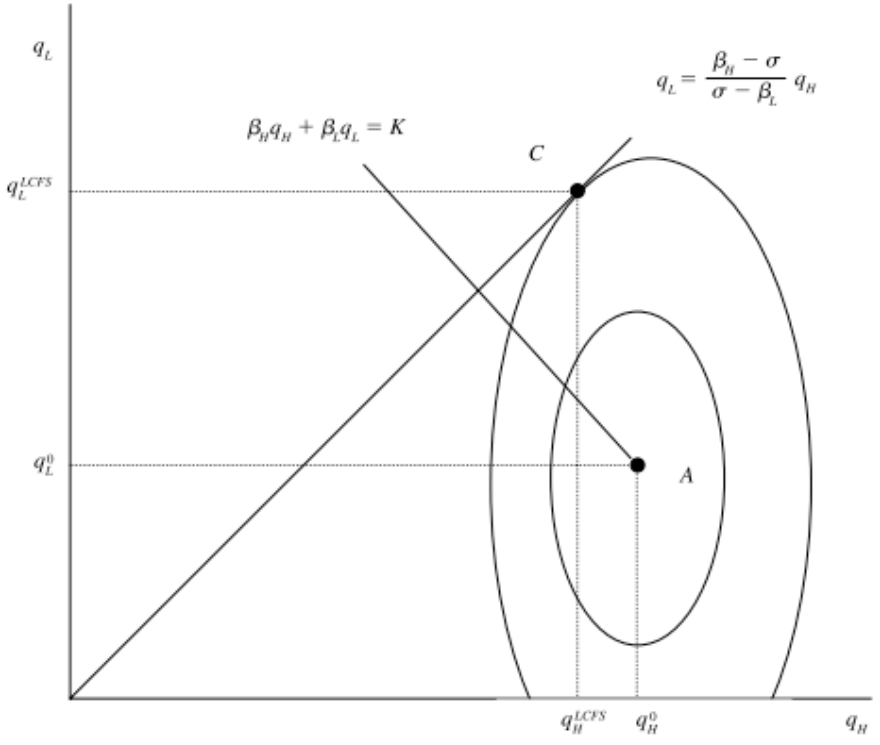


Intensity standards: intuition

Two forces are operating at once:

1. **Composition effect:** shifts production toward lower-emissions output
2. **Scale effect:** implicit output subsidy encourages more production

If the scale effect dominates, total emissions can rise even as emissions per unit fall



Second-best: distribution-constrained policy

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A planner that values equity places weight $\lambda > 0$ on this burden

Distribution-constrained optimum

Planner problem:

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$$\tau^{SB} < d \quad \text{whenever } \lambda \kappa > 0$$

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This is a sharp second-best result: distribution constraints can lower the optimal carbon price

Revenue recycling and the constraint

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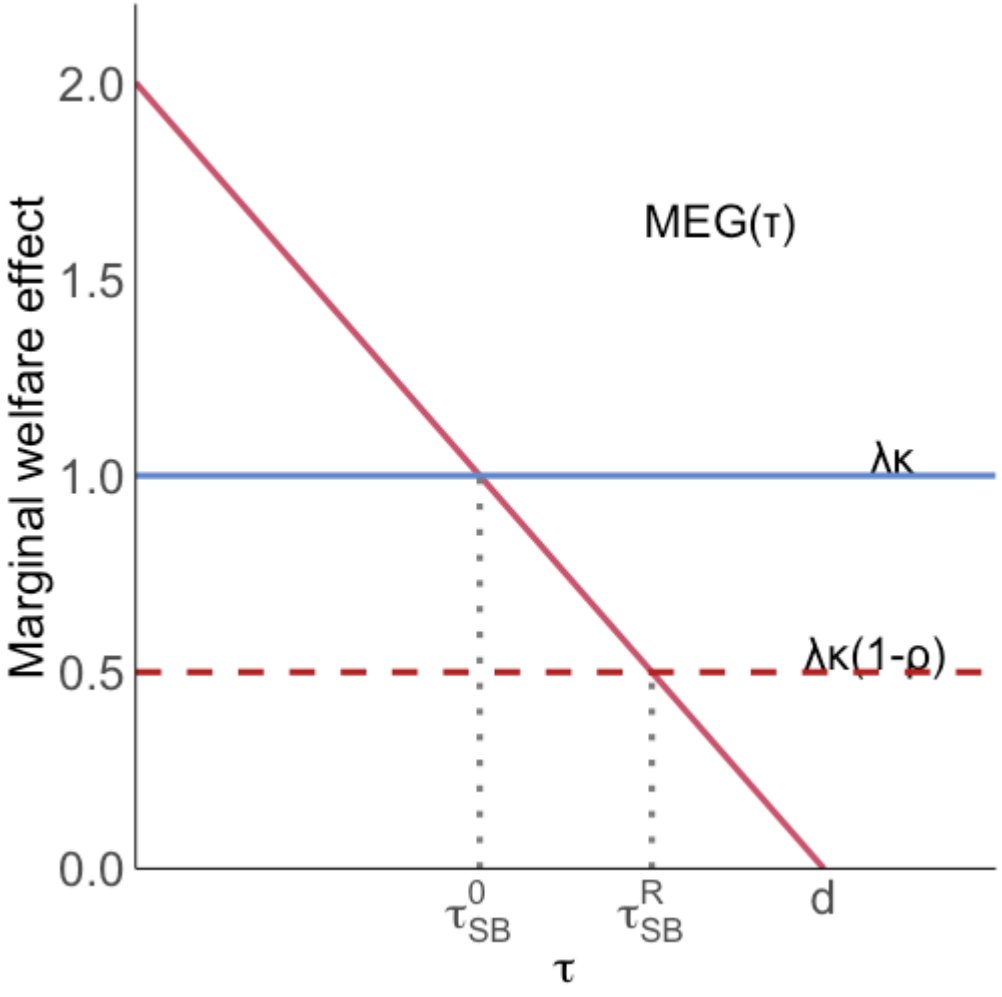
Higher targeted recycling ($\rho \uparrow$) reduces the equity wedge and raises the feasible optimal carbon price toward d

Distribution-constrained: graphical



The red line is marginal efficiency gain from a higher carbon price

Horizontal lines are marginal distributional costs

Targeted recycling lowers the distributional cost curve and moves τ^{SB} closer to d

Distribution takeaway

In a second-best world, we should evaluate policy bundles:

$$\{\text{carbon price},\ \text{recycling design}\}$$

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Takeaway: distributional design is not an add-on; it is part of optimal second-best climate policy

Dynamic second-best: innovation spillovers

Climate policy is also dynamic: today's innovation changes tomorrow's abatement costs

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A carbon price targets the emissions externality

But clean innovation can generate knowledge spillovers that firms cannot fully appropriate

So even with a correctly-set carbon price, private R&D can still be too low

Dynamic model: private conditions

Let firms choose emissions E and clean R&D investment I

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If firms capture only a share $\eta \in (0,1]$ of innovation benefits, private FOCs are:

$$-C_E(E,\theta) = \tau$$

$$(1-s)\Psi'(I) = \eta\left[-C_\theta(E,\theta)\right]$$

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Social planner:

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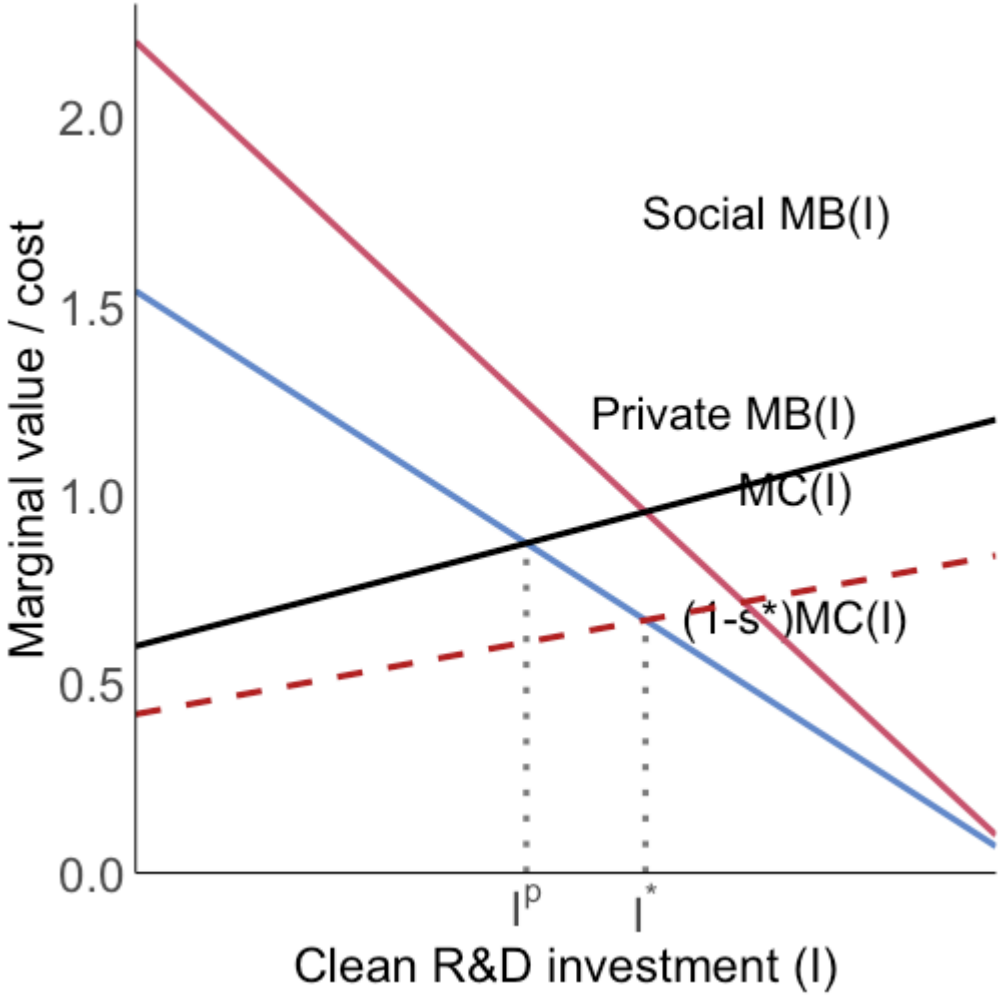
If $\eta < 1$, we need a positive R&D subsidy in addition to the carbon price

Dynamic second-best: graphical



Without subsidy, private R&D is I^p where private MB crosses MC

With spillovers ($\eta < 1$), this is below the social optimum I^*

An R&D subsidy lowers effective marginal cost to $(1 - s^*)MC$, moving investment toward I^*

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Second-best takeaway: dynamic efficiency typically requires pricing emissions and subsidizing clean innovation together